

Title:Bank Loan Analysis Dashboard using Power BI

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SYNTEXHUB INTERN

1. Introduction

The banking sector handles large amounts of loan-related data every day. Analyzing this data helps financial institutions understand customer behavior, loan risks, repayment patterns, and approval factors. This project focuses on analyzing bank loan data using Power BI to generate meaningful insights and interactive visualizations.

The dashboard helps identify:

Good and bad loans

Customer financial behavior

Factors affecting repayment

Loan distribution patterns

Credit risk indicators

The project demonstrates the use of:

Data cleaning

Data visualization

KPI analysis

Trend analysis

Business intelligence techniques

2. Objectives of the Project

The main objectives of this project are:

To analyze bank loan data effectively.

To classify good and bad loans.

To identify factors affecting loan approval and repayment.

To perform trend analysis using customer and financial data.

To create an interactive Power BI dashboard.

To generate business insights for decision-making.

3. Tools and Technologies Used

Tool	Purpose
Power BI	Dashboard creation and visualization
Excel/CSV Dataset	Data source

DAX KPI calculations

Power Query Data cleaning and preprocessing

4. Dataset Description

The dataset contains customer and loan-related information used for loan analysis.

Important Columns Used

Column Name

Description

credit_score

Customer credit score

loan_amnt

Loan amount

loan_int_rate

Loan interest rate

loan_status

Good or bad loan

loan_intent

Purpose of loan

person_income

Customer income

person_age

Customer age

person_gender

Gender

person_education

Educational qualification

person_emp_exp

Employment experience

person_home_ownership

Home ownership status

5. Dashboard Visualizations

The dashboard includes multiple interactive visualizations.

5. 1KPI Cards

The following KPI cards were created:

Total Applications

Total Loan Amount

Average Loan Amount

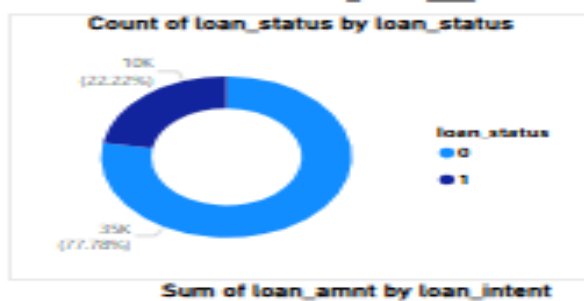
Average Interest Rate

Good Loans

Bad Loans

These KPIs provide a quick overview of loan performance.

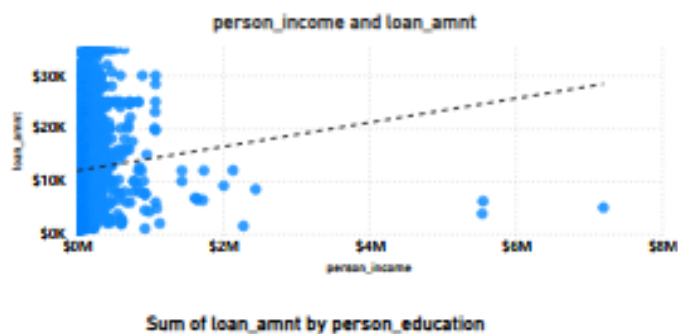
5.2 GRAPHS



Good vs Bad Loans (Donut Chart)

Insight

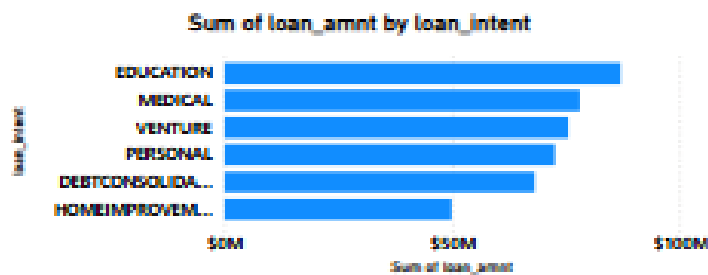
- 1.The majority of loans are classified as bad loans.
- 2.Only a smaller percentage represents good repayment behavior.
- 3.Loan default risk is relatively high in the dataset.



Income vs Loan Amount (Scatter Plot)

Insight

1. Customers with higher income generally receive larger loan amounts.
2. Most customers are concentrated in lower-to-medium income ranges.
3. The trend line shows a positive relationship between income and loan amount.



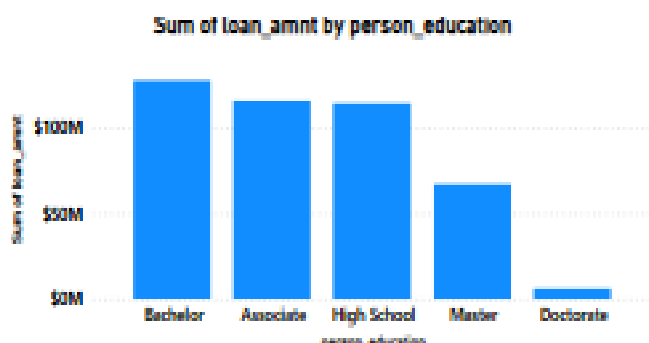
Loan Amount by Loan Intent (Bar Chart)

Insight

Education loans have the highest loan amount.

Medical and venture-related loans also contribute significantly.

Personal and home improvement loans are comparatively lower.



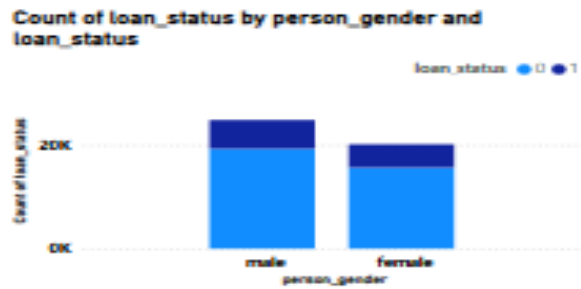
Loan Amount by Education (Column Chart)

Insight

Customers with Bachelor's and Associate education levels receive the highest loan amounts.

Doctorate holders represent the lowest loan distribution.

Educational background influences loan demand and approval.



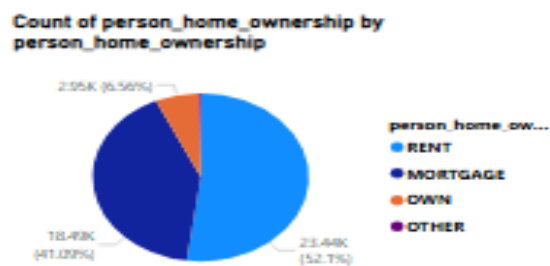
. Loan Status by Gender

Insight

Male customers contribute more loan applications than female customers.

Both genders contain good and bad loans, but bad loans dominate overall.

Gender-based repayment patterns can be analyzed further for risk assessment.



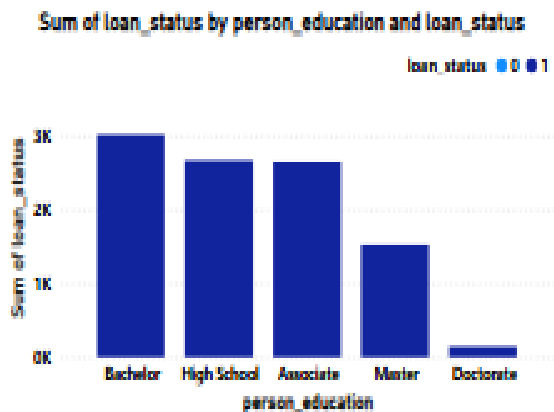
Home Ownership Analysis (Pie Chart)

Insight

Most customers belong to rented or mortgage housing categories.

Customers with owned homes are fewer but may indicate stronger financial security.

Home ownership status can influence loan repayment capability.



Education vs Loan Status

Insight

Bachelor and High School education groups contain the largest number of loans.

Higher education groups appear to have comparatively fewer defaults.

Educational qualification impacts financial stability and repayment behavior.



Credit Score vs Loan Status

Insight

Customers with higher credit scores show better loan quality.

Lower credit score ranges are associated with increased bad loans.

Credit score is one of the strongest indicators of repayment behavior and approval risk.

Key Insights

Most loans are classified as good loans.

Customers with high income tend to receive larger loan amounts.

Credit score strongly affects loan quality and repayment behavior.

Educational background influences financial stability.

Home ownership positively impacts repayment capability.

Higher interest rates are associated with higher loan risk.

Loan intent categories such as education and medical purposes have high loan demand.

6.Conclusion

The Bank Loan Analysis Dashboard provides valuable insights into customer financial behavior, loan distribution, repayment trends, and risk factors. The dashboard helps identify important factors affecting loan approval and repayment performance.

Using Power BI visualizations and DAX measures, the project successfully transformed raw loan data into meaningful business intelligence. The analysis can support financial institutions in improving loan decision-making, risk assessment, and customer evaluation processes.

This project demonstrates practical skills in:

Data analysis

Dashboard development

Business intelligence

Data visualization

Financial analytics

7.Future Improvements

Future enhancements may include:

Real-time loan monitoring

Predictive loan default analysis

Machine learning integration

Regional and time-based analysis

Advanced customer segmentation